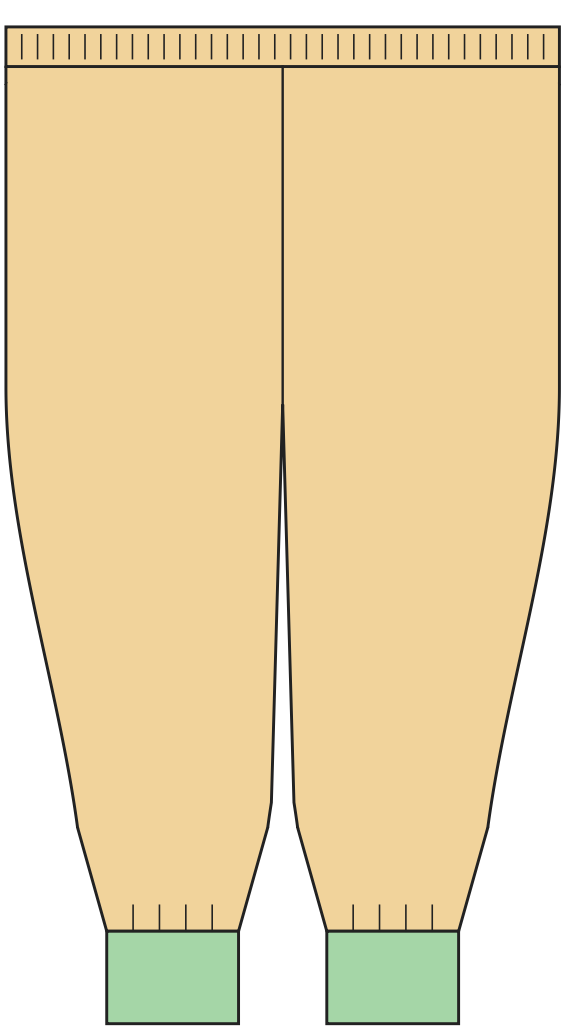
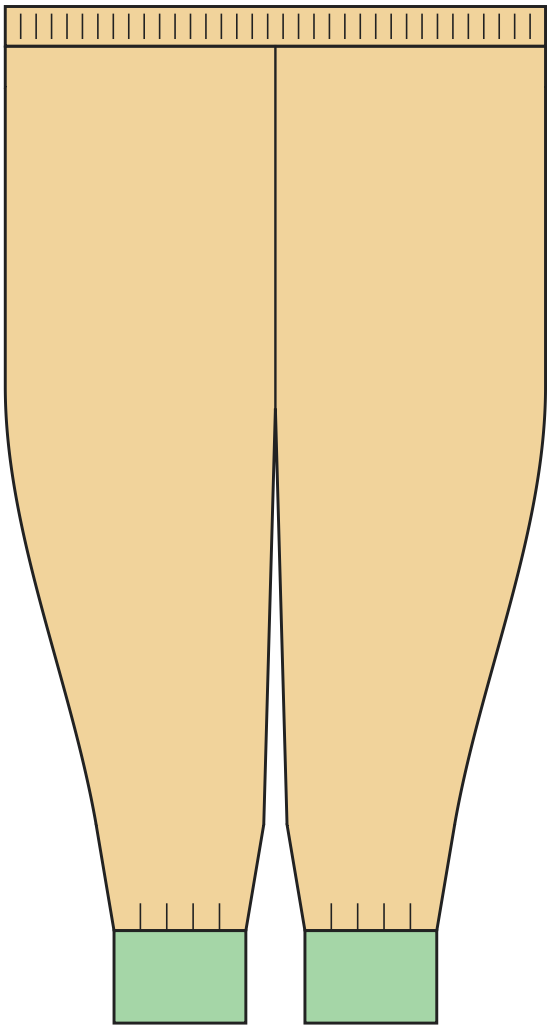
Harem pants

size 122 · harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch)

4 parts · 20 steps · medium · about 3 h



Front view



Back view

Sewn with: Sewing machine



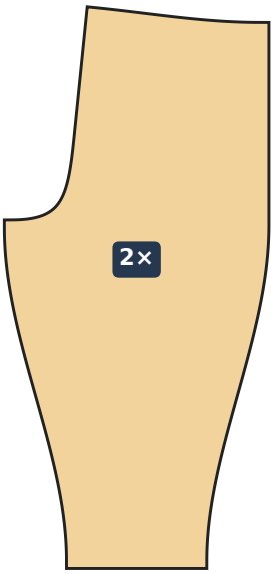
Sewing machine



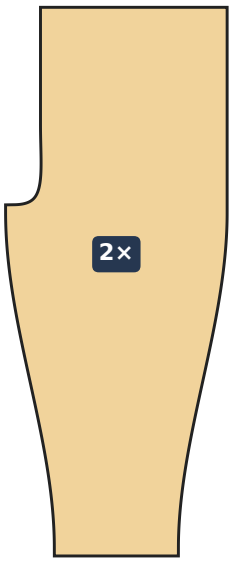
Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

Parts



B · Back leg



A · Front leg



D · Leg cuff



C · Waistband (elastic) (cut on fold)

Seam allowance: included: 1.0 cm · hem 1.0 cm

Material

Main fabric
0.90 m at 150 cm width

Rib knit
0.25 m at 70 cm width

Elastic
Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Sewing machine
narrow zigzag 1 × 2.5 mm (or overlocker); wide zigzag 4 × 2.5 mm; twin needle 4 mm, stitch 3 mm

Iron
steam

Polyester sewing thread
In the colour of the fabric.

Pins or clips
Pin across the seam line.

Chalk or a fabric pen
For notches, labels and placement lines.

Colours in this booklet

Main fabric
right side · wrong side

Rib knit
right side · wrong side

Sew now (machine)

Sew by hand

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

Basics: Sewing machine

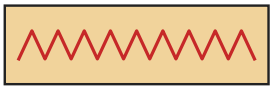
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

Basics: the sewing machine



Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



Zigzag

Knit seams (narrow) and finishing the edges (wide).

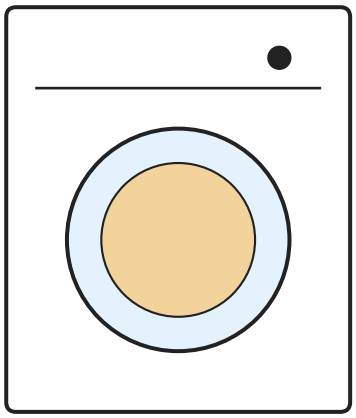


Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



30–40°

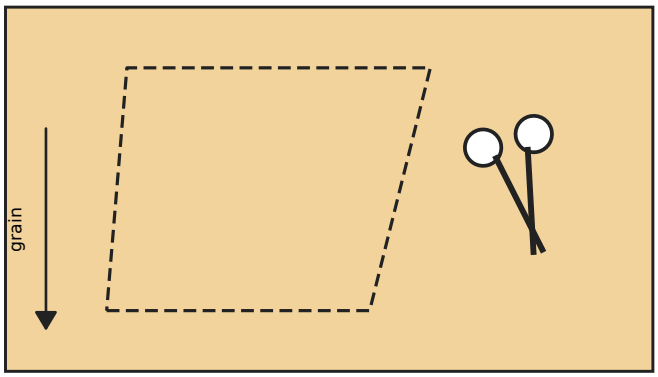
2 Cut out all pieces



Scissors

1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.

Tip: Cut both layers at once with long strokes; keep the fabric flat on the table.

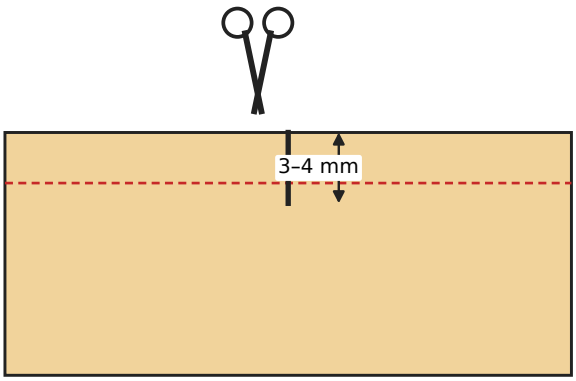


3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

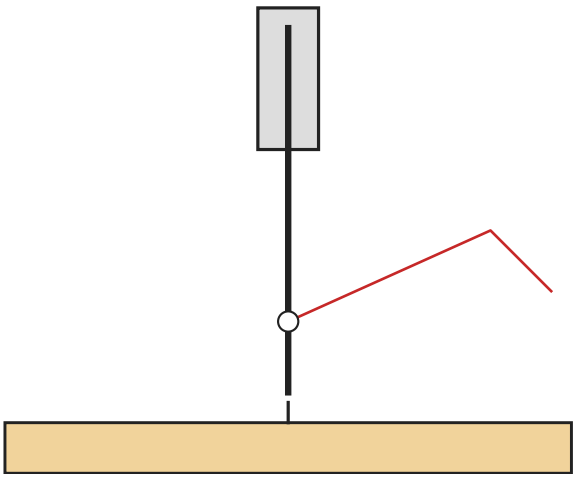
⚠ Important: Never deeper than half the seam allowance.



4 Set up the machine

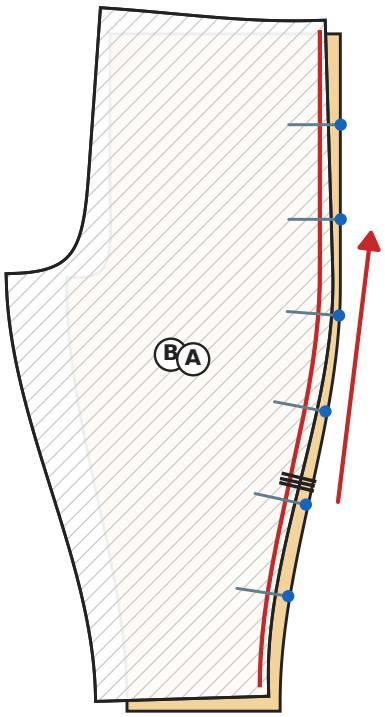
1. Jersey/stretch needle 75–80, polyester thread.
2. Sew a test seam on a scrap first.

Tip: See the basics page at the front of the booklet.



5 Sew the side seams

Sewing machine · Pins



1. Front and back leg right sides together
2. Single notches = front, double = back.

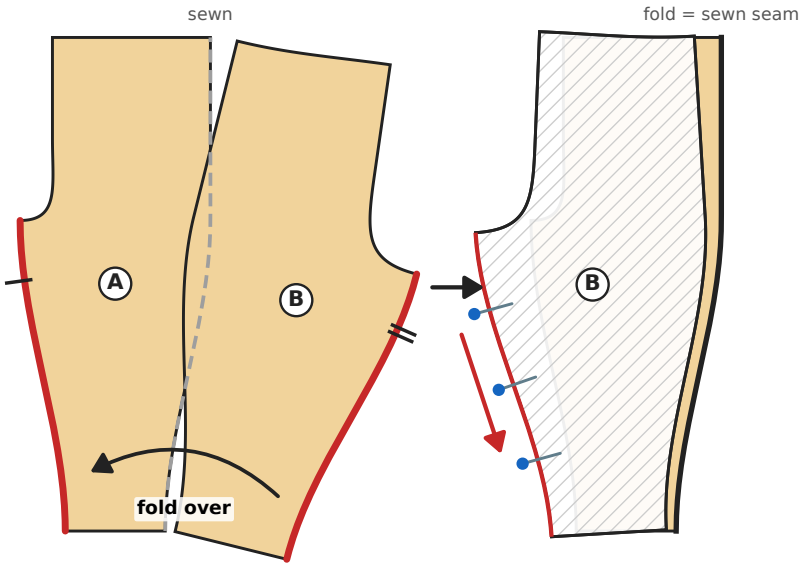
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Always the front and back of the SAME leg (FL + BL, FR + BR).

2×

6 Sew the inseams

Sewing machine · Pins



1. The notches on the inseam meet.

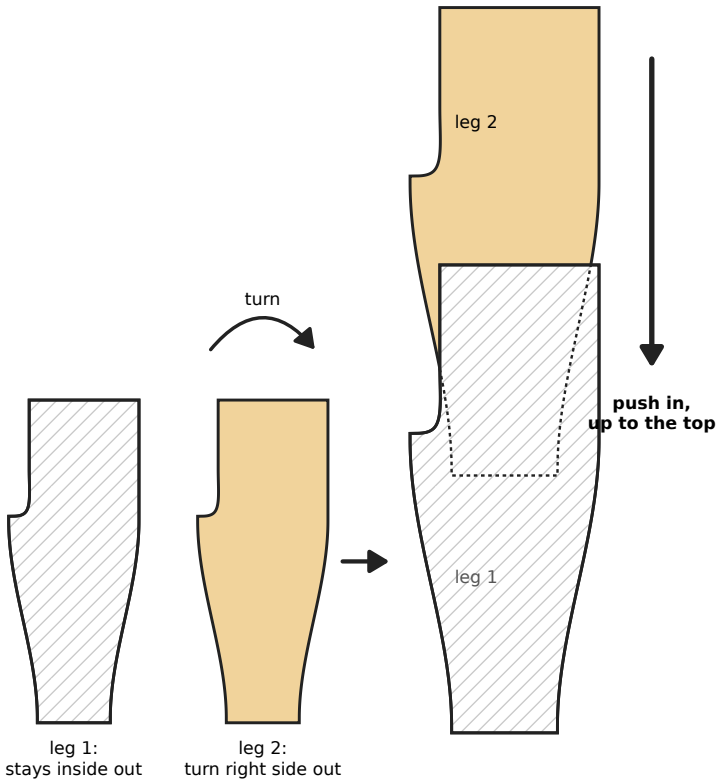
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Start exactly at the crotch point and sew towards the hem.

Tip: Pin the notches first, then the ends.

2×

7 Put one leg inside the other

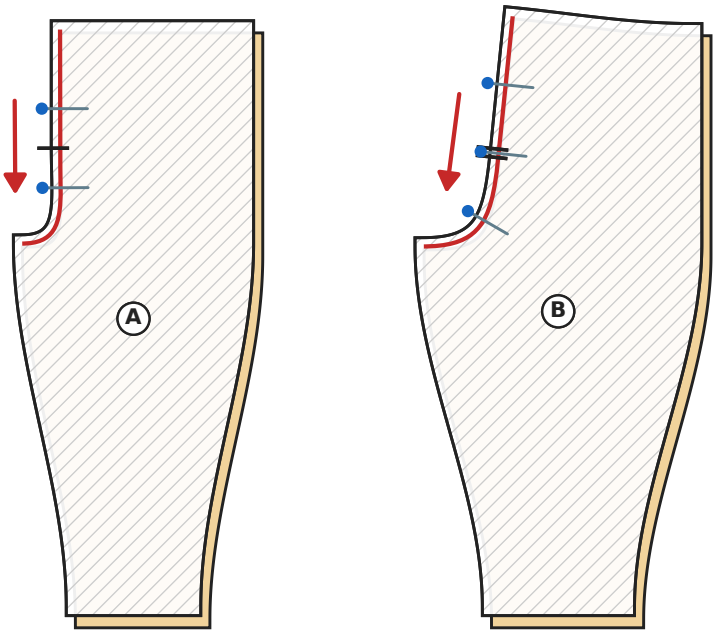


1. Turn leg 2 right side out.
2. Leg 1 stays wrong side out.
3. Push leg 2 into leg 1 until the top edges are even
4. The hems point the same way.

Tip: Now the legs lie right sides together and the crotch curves lie exactly on top of each other.

8 Sew the crotch seam





1. In one go from the front to the back top edge
2. The inseams meet exactly.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Sew the crotch curve a second time for strength.

Tip: The inseams lie exactly on top of each other; press their allowances in opposite directions. Pull the inner leg out afterwards.

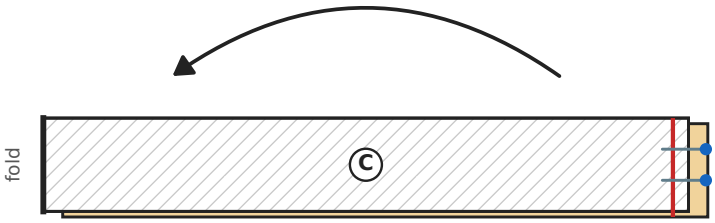
9 Close the waistband into a ring

Sewing machine

Pins

1. Right sides together across the short ends (centre back)
2. Leave 3 cm open for the elastic.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

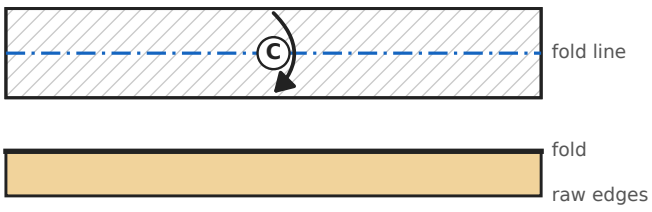


10 Fold the waistband in half

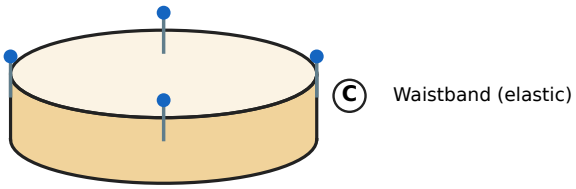
Iron

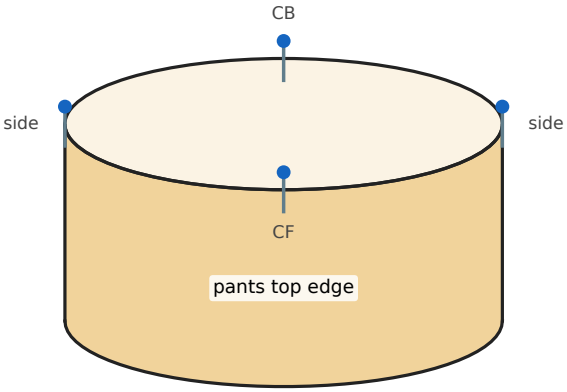
1. Lengthwise, wrong sides together, along the fold line.

Iron: steam



11 Mark quarters





1. On waistband and pants top edge: centre front, centre back, sides.

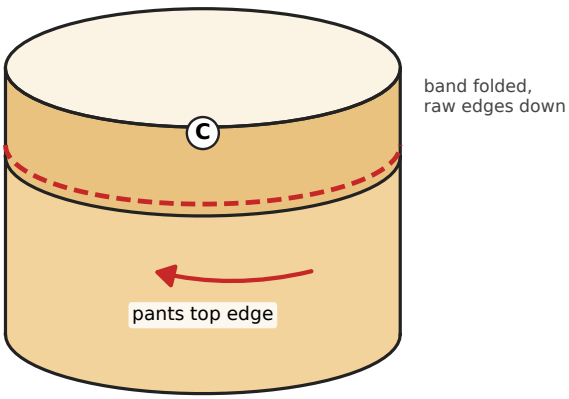
12 Sew the waistband on


Sewing machine


Pins

1. Raw edges together, opening at the back on the inside
2. Quarter marks meet.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



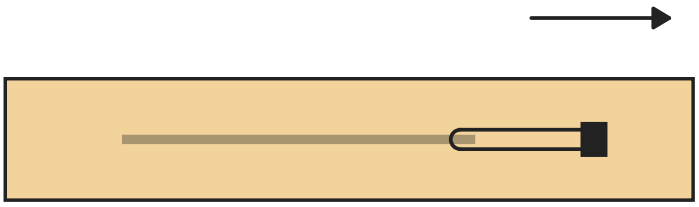
13 Thread the elastic

Sewing machine

Safety pin

1. With a safety pin through the opening
2. Overlap the ends by 2 cm and sew them together.

Sewing machine: wide zigzag 4 × 2.5 mm



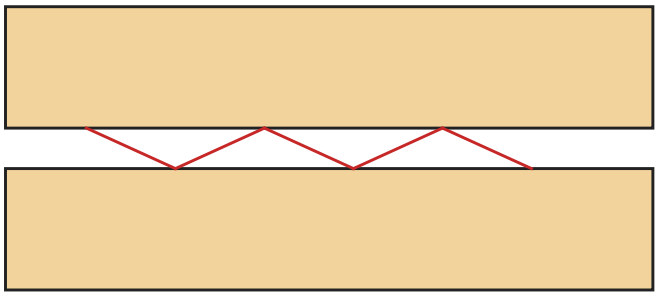
elastic, with a safety pin

14 Close the opening

Sewing machine

1. Spread the gathers evenly, close the opening and stitch in the ditch at the side seams so the elastic cannot twist.

Sewing machine: twin needle 4 mm, stitch 3 mm



ladder stitch: the stitches disappear

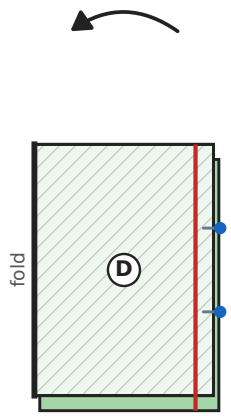
15 Close the cuffs into rings

Sewing machine

Pins

1. Right sides together across the short ends.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



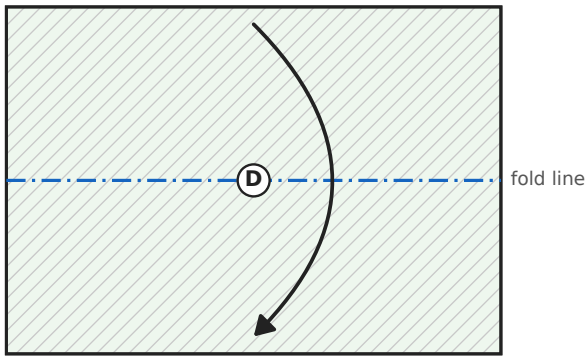
2×

16 Fold the cuffs in half

Iron

1. Lengthwise, wrong sides together.

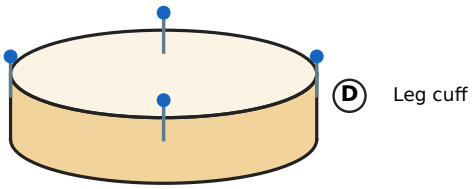
Iron: steam

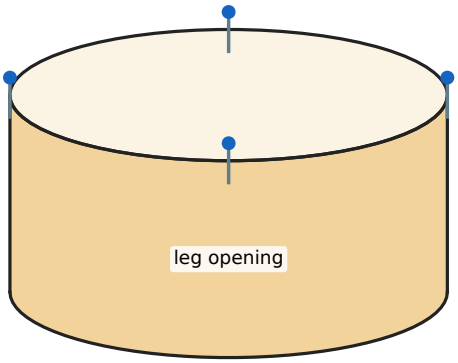




2×

17 Mark quarters





1. On cuff and leg opening.

2×

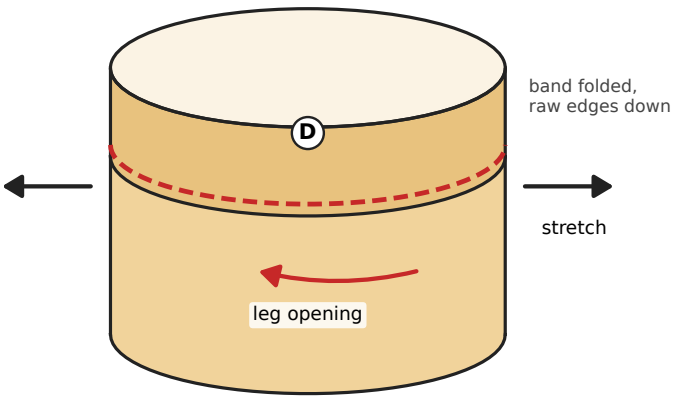
18 Sew the cuffs on


Sewing machine


Pins

1. Slip the cuff over the right-side-out leg, cuff seam at the inseam.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



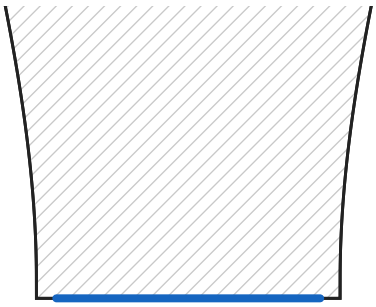
2×

19 Press the allowance up

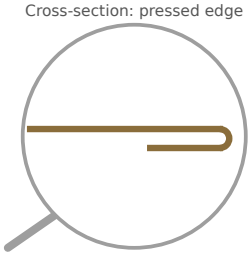
Iron

1. Seam allowance towards the pants.

Iron: steam



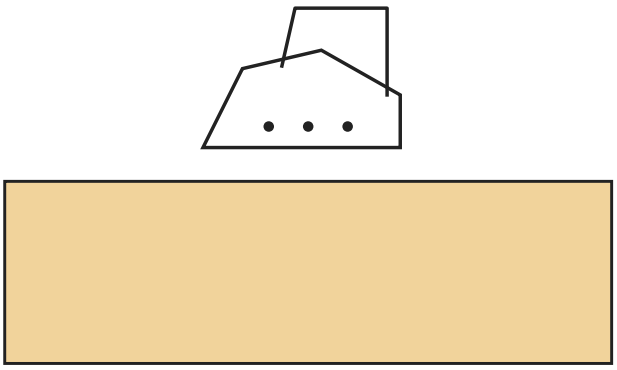
Cross-section: pressed edge



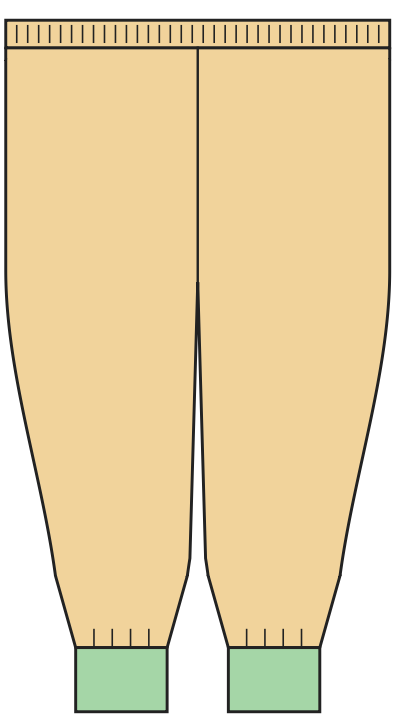
2×

20 Final pressing

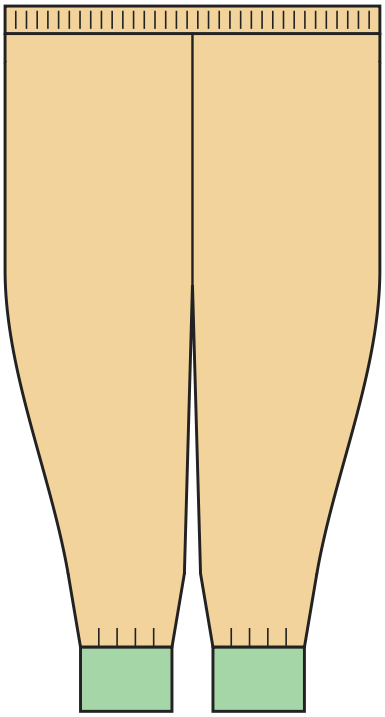
1. Press all seams with steam
2. Do not press down hard on sweat.



Done!



Front view



Back view

After the test garment: Try the test garment on the child: length, width at the hip and the waistband. Adjust the measurements or options and print the pattern again before you cut the good fabric.

Harem pants · size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (12 pages + overview): `harem_pants_size122_a4_tiled.pdf`
- Preview A4 tiles: `harem_pants_size122_a4_preview.png`
- SVG master 1:1: `harem_pants_size122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	122.0 cm	table
Chest	62.0 cm	table
Waist	56.0 cm	table
Hip	66.0 cm	table
Head	52.0 cm	table
Neck	28.5 cm	table
Inseam	54.0 cm	table
Side length	72.0 cm	table
Arm length	47.0 cm	table
Back length	29.5 cm	table
Shoulder width	10.0 cm	table
Thigh	37.0 cm	table
Ankle	17.8 cm	table

Finished measurements (virtually sewn)

Finished measurement	Value
Hip width	85.8 cm
Thigh width	55.5 cm
Leg width at mid-leg	42.0 cm
Leg opening	20.0 cm
Waistband relaxed	50.4 cm
Waistband stretched (pants top edge)	83.0 cm
Outseam	75.5 cm
Inseam	48.2 cm
Front rise	27.3 cm
Back rise	29.1 cm
Waistband height	3.0 cm
Cuff height	7.0 cm

Finished measurement	Value
Crotch seam total	64.2 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Back leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Waistband (elastic)	cut 1 on fold	Main fabric	seam allowance 1.0 cm included
Leg cuff	cut 2	Rib knit	seam allowance 1.0 cm included

Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.25 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	66.0 cm	66.0 cm	+0.0 mm	□ OK	
Seam pairs	Inseam front ↔ back	41.7 cm	41.7 cm	+0.0 mm	□ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	26.9 cm	26.9 cm	+0.0 mm	□ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	31.3 cm	31.3 cm	+0.0 mm	□ OK	
Seam pairs	Waistband ↔ pants top edge	83.0 cm	83.0 cm	+0.0 mm	□ OK	
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 28.5 cm	20.0 cm = 70 %	-85.5 mm	□ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	□ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	□ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.9°	+0.9°	□ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.1°	+0.1°	□ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	□ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.2°	+0.2°	□ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.2°	+0.2°	□ OK	

Area	Check	Target	Actual	Δ	Result	Note
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	□ OK	
Angles/transitions	Side seam ↔ waistband (front + back)	180°	180.1°	+0.1°	□ OK	
Measurements	Hip circumference pattern = body + ease	85.8 cm	85.8 cm	+0.0 %	□ OK	body 66.0 cm + +30 % ease
Measurements	Top edge fits over the hip	≥ 67.3 cm	95.4 cm	+281.1 mm	□ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	14.6 cm – 31.4 cm	27.3 cm	+0.0 mm	□ OK	body rise 18.0 cm, harem drop 7.3 cm
Measurements	Inseam length plausible	39.0 cm – 60.5 cm	55.5 cm	+0.0 mm	□ OK	body inseam 54.0 cm
Measurements	Foot fits through the leg opening	≥ 24.1 cm	29.5 cm	+54.8 mm	□ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		□ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		□ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		□ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		□ OK	
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		□ OK	

Result: 31 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 858}}`).

Name	Value	Derived from	Note
`hip_width`	85.8 cm	hip × (1 + 30 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	20.9 cm	hip_width / 4 − front_back_offset	
`back_width`	21.9 cm	hip_width / 4 + front_back_offset	
`front_crotch_extension`	4.3 cm	5 % of hip_width	

Name	Value	Derived from	Note
`back_crotch_extension`	8.6 cm	10 % of hip_width	
`rise`	24.3 cm	(side length − inseam) + seat ease + harem drop − waistband height	
`inseam_finished`	48.2 cm	inseam − allowance to the floor − harem drop + blousing	
`hem_width`	28.5 cm	cuff length / cuff stretch ratio	

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